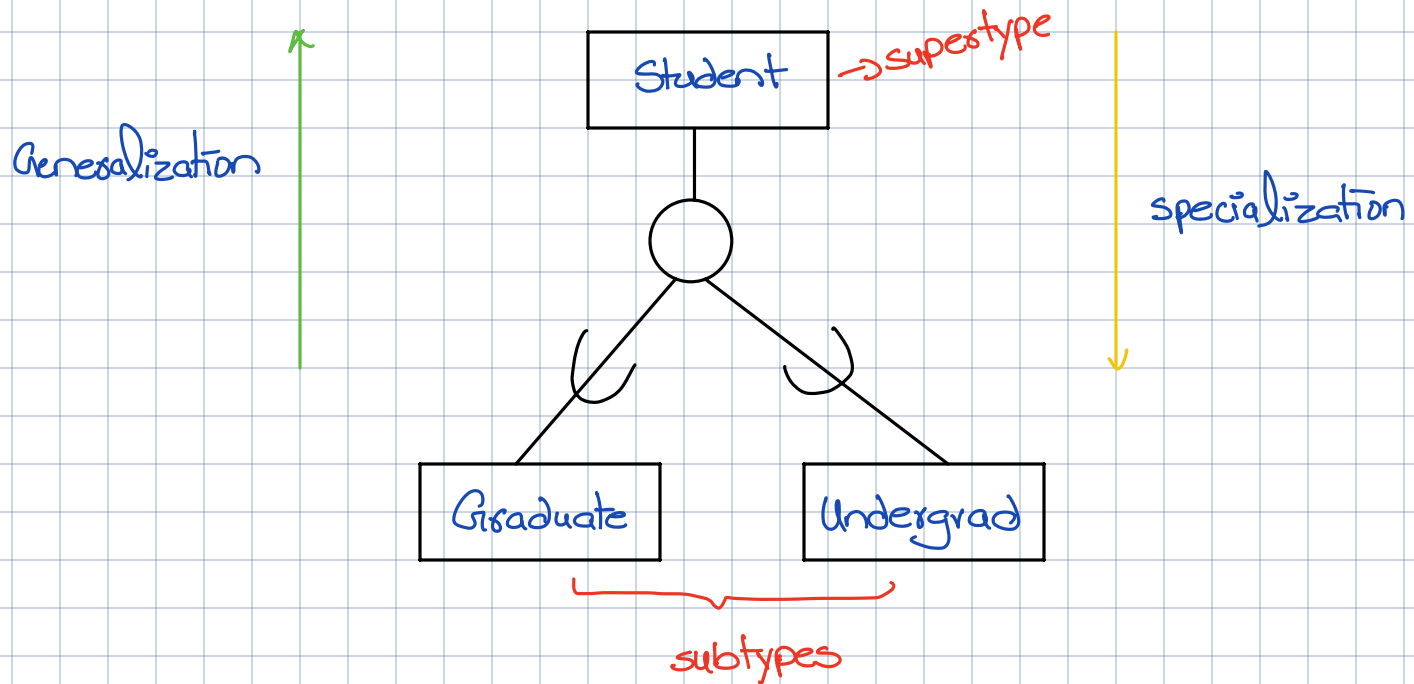


Enhanced ER Model:

↳ Data is more complex now.

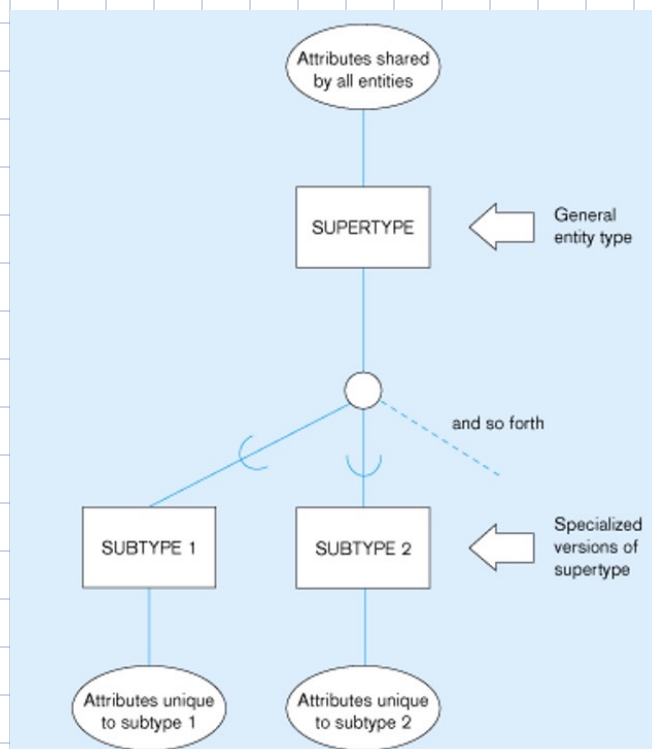
↳ EER improves flexibility of the basic E-R Model to represent business data.

Entity Type hierarchies:

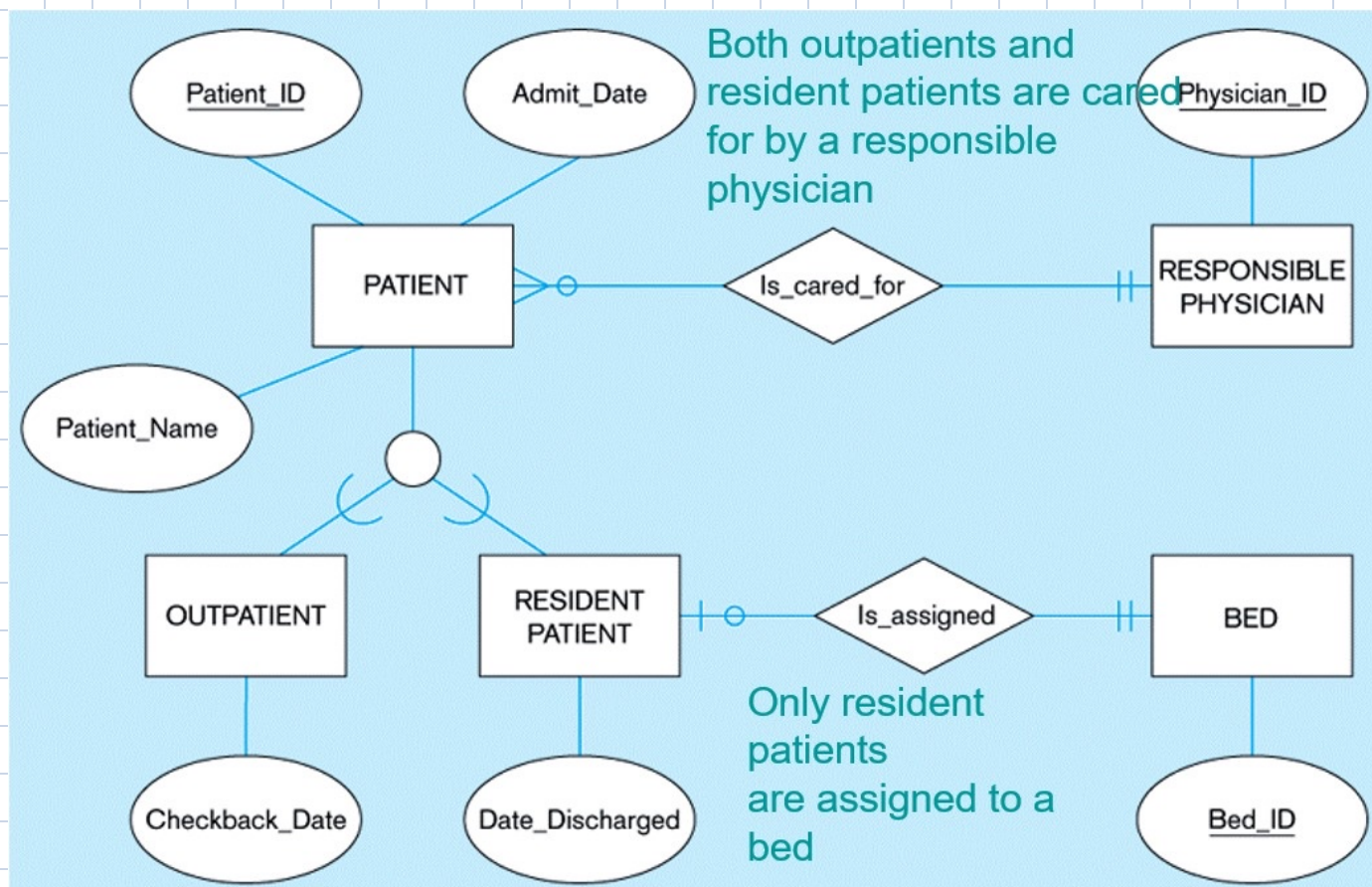


↳ There is-A relation b/w subtype & supertype.

↳ SubType inherit attributes of the supertype.



↳ Relationship with supertype also extends to its subtypes but not viceversa.



ER Constraints:

1. Completeness:

↳ Whether an instance of a supertype must also be a member of at least one subtype?

for example,

If students (supertype) has only grad & ug (subtypes).

↳ Total specialization → Yes (Add double line)

↳ Partial specialization → No (Leave single line)

2. Disjoint constraints:

↳ Whether an instance of a supertype may simultaneously be a member of two or more subtypes?

↳ Ahsan is both faculty & Alumnus

↳ Disjoint → No (Add letter 'd')

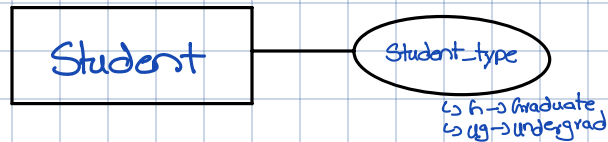
↳ Overlap → Yes (Add letter 'o')

3. Subtype discriminators.

↳ Place an attribute in supertype whose values determine the type of subtype(s).

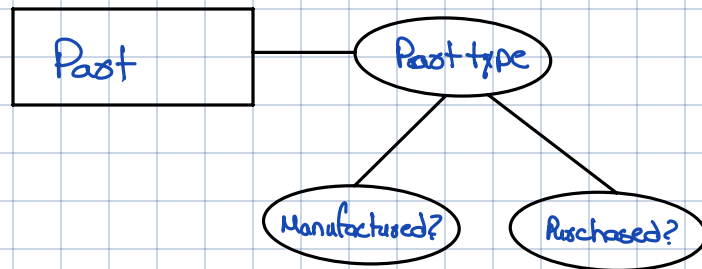
↳ Disjoint:

↳ A simple attribute with values for possible subtype.



↳ Overlap

↳ A composite attribute whose subparts pertain to different subtypes



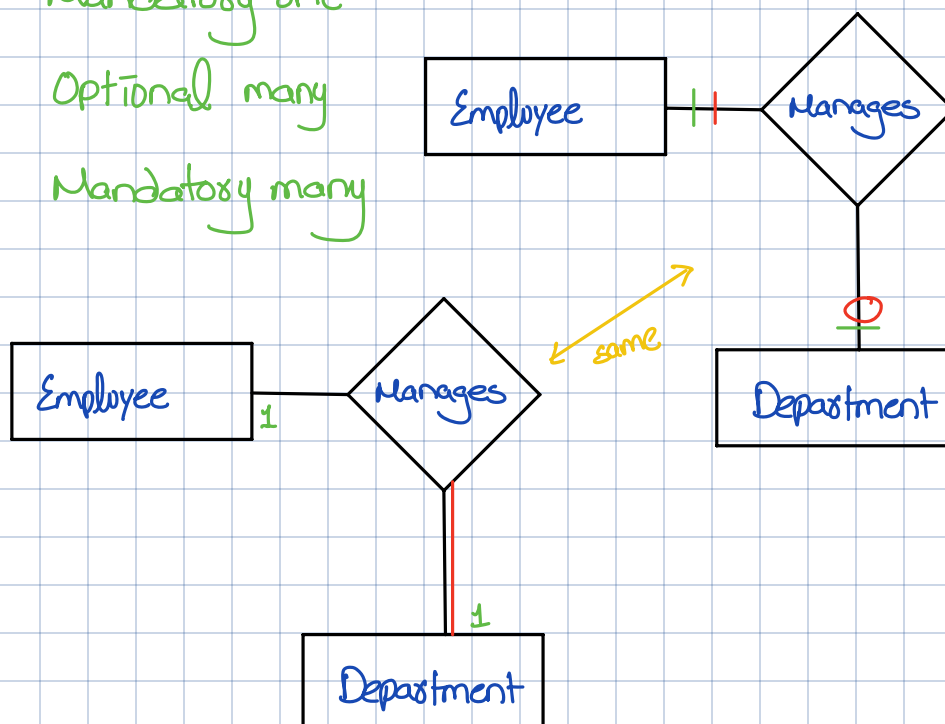
Crow-foot Notation:

0 1 Optional one

1 1 Mandatory one

0 E Optional many

1 E Mandatory many



PRACTICE:

